

Optimal Credit Limit Extension Policy Using Multi-Period Lifecycle Simulation

A Risk-Adjusted Profitability Framework for Consumer Lending

Prepared by: Praveen Kumar , praveenkm31@gmail.com

+918585010275

<https://www.linkedin.com/in/praveenk-8/>

| Branch: term_ext_lifecycle

1. Executive Summary

This report presents a data-driven framework for determining the optimal credit limit extension policy across a 30,000-customer consumer lending portfolio. Rather than treating the extension decision as a one-time event, the analysis models customers dynamically over a 12-month horizon, re-evaluating each account monthly as their financial behaviour evolves.

96.1% of customers received at least one offer; 83.8% of Fair-tier customers improved to Good or Excellent by year-end

Key findings at a glance:

- Net profit over 12 months: \$1.41M (Revenue \$2.12M, Loss \$0.71M) — 66.6% margin
- Total extensions granted: 70,317 across 12 monthly campaigns
- Portfolio default rate: 0.66% — 7.5× below the 5% LP risk cap
- 96.1% of customers received at least one offer; 83.8% of Fair-tier customers improved to Good or Excellent by year-end
- Primary profit lever: raising the extension fee from \$40 to \$48 adds ~\$180K with zero additional risk
- Primary risk lever: reducing eligibility to Excellent/Good customers in early campaigns contains downside



Figure 1: Executive Dashboard — KPIs, monthly P&L, segment profit, risk-return scatter

2. Problem Statement & Objectives

A consumer lender holds a portfolio of 30,000 accounts. Each account is eligible, under certain conditions, to receive a term extension on their existing loan — effectively extending the repayment window in exchange for a fee. The business problem is: who should receive an offer, at what fee, and when?

This is not a trivial decision. Offering extensions too broadly increases default exposure. Offering too conservatively leaves revenue on the table. A static, one-time policy cannot adapt to how customers behave over time — a borrower who repays on time in month one is fundamentally less risky in month two than a borrower who does not.

The analysis addresses five core questions:

- Q1: What is the optimal extension policy that balances profitability and risk?
- Q2: What operations research techniques best model this multi-period decision?
- Q3: How does borrower behaviour — repayment, churn, acceptance — impact the optimal strategy?
- Q4: How do external economic conditions (GDP, unemployment, interest rates) affect outcomes?
- Q5: What levers — fee pricing, eligibility gates, risk concentration limits — maximise value?

3. Methodology

3.1 Data & Feature Engineering

The analysis uses a 30,000-account dataset with six features: initial loan balance, days since last loan disbursement, on-time payment percentage, number of prior increases in 2023, and cumulative profit contribution. A Borda-count composite credit score is computed as a weighted percentile rank across these features:

$$\begin{aligned} \text{credit_score} = & 0.35 \times \text{rank}(\text{on_time_pct}) + 0.30 \times (1 - \text{rank}(\text{num_increases})) \\ & + 0.20 \times (1 - \text{rank}(\text{days_since})) + 0.15 \times \text{rank}(\text{profit_contribution}) \end{aligned}$$

K-Means clustering (k=4) on the risk rank assigns each customer to one of four states: Excellent, Good, Fair, or Poor. These states drive all downstream parameters.

3.2 Macro Enrichment

Live macroeconomic data is pulled from the FRED API (GDP, unemployment rate, federal funds rate, CPI). In the absence of live data, a calibrated baseline is used. Macro conditions adjust both the hazard rate (default probability) and the available extension term days, creating three scenario paths: optimistic, baseline, and adverse.

3.3 Monte Carlo Term Simulation

For each eligible customer in each campaign, a vectorised Monte Carlo simulation draws 200 scenarios of the extension term (T, from a lognormal distribution clipped to 30–180 days) and computes the expected revenue and expected loss under each:

$$\begin{aligned} \text{Revenue} &= \$40 \times \text{annuity_factor}(T, r=19\%) \\ \text{Loss} &= \text{Balance} \times \text{LGD} \times p_default(T) \times \exp(-r \times T / 730) \\ p_default(T) &= 1 - \exp(-\lambda \times T / 365) \end{aligned}$$

The per-state hazard rates (λ) are: Excellent 1.5%, Good 4.0%, Fair 10.0%, Poor 22.0%. Loss Given Default (LGD) is 60% — consistent with industry norms for unsecured consumer credit.

3.4 GBM Uptake Model

Customer acceptance of the extension offer is modelled via Geometric Brownian Motion, calibrated per credit state. The base acceptance rates are Excellent 80%, Good 70%, Fair 55%, Poor 35% — reflecting lower intent among higher-risk borrowers. In the lifecycle model, an explicit Bernoulli draw is made for each offered customer, unlike the one-shot model where uptake only weights the LP objective.

3.5 Linear Programme Optimisation

A binary LP determines which eligible customers receive an extension offer each campaign. The objective maximises total expected profit, subject to two hard constraints:

$$\begin{aligned} \max \quad & \sum \text{profitability_score}_i \times \text{uptake}_i \times x_i \\ \text{s.t.} \quad & \sum (p_{\text{default}_i} - 0.05) \times \text{balance}_i \times x_i \leq 0 \quad [5\% \text{ portfolio risk cap}] \\ & \sum \text{balance}_i \times x_i \leq \$500,000,000 \quad [\text{exposure cap}] \\ & x_i \in \{0, 1\} \end{aligned}$$

The solver (SciPy HiGHS) runs on the eligible customer subset each campaign. Ineligible customers (those with an active extension or fewer than 60 days since their last loan) are excluded automatically.

3.6 Multi-Period Lifecycle Simulation

The core innovation of this framework is the campaign loop. Every 30 days, the system: (1) completes finishing extensions by drawing actual outcomes — early, on-time, or default; (2) updates each customer's features based on that outcome; (3) re-scores their credit state against the frozen t=0 reference population; (4) checks eligibility; and (5) runs the full MC → GBM → LP pipeline on those now eligible. This creates a dynamic, self-correcting policy where customers who demonstrate good repayment behaviour graduate to better credit states and receive more favourable terms in subsequent campaigns.

4. Mathematical Formulation

The problem is a finite-horizon, discrete-time Markov decision process, solved approximately per period via a binary linear programme. The full formulation is as follows.

4.1 State Space

Each customer i at time t is characterised by state vector $S_i(t)$:

$$S_i(t) = (\text{credit_state}_i(t), \text{on_time_pct}_i(t), \text{balance}_i, \text{days_since}_i(t))$$

Credit state $\in \{\text{Excellent}, \text{Good}, \text{Fair}, \text{Poor}\}$, assigned by K-Means on risk_rank.

4.2 Transition Dynamics

After extension outcome $o \in \{\text{on-time}, \text{early}, \text{default}\}$, features update deterministically:

$$\text{on_time_pct}_i(t+1) = \text{clip}(\text{on_time_pct}_i(t) + \Delta_o, 0, 100)$$

$$\Delta_{\{\text{on-time}\}} = +2pp, \Delta_{\{\text{early}\}} = +1pp, \Delta_{\{\text{default}\}} = -10pp$$

The new credit state is derived by re-computing the Borda-count score with updated features and assigning the nearest K-Means centroid (centroids frozen at $t=0$).

4.3 Optimisation per Period

At each campaign t , the LP solves:

$$\text{Objective: } \max \sum_i \pi_i(t) \cdot u_i(t) \cdot x_i$$

$$\text{where: } \pi_i(t) = E[\text{Revenue}_i] - E[\text{Loss}_i] \quad (\text{profitability score})$$

$$u_i(t) = \text{GBM uptake probability}$$

$$x_i \in \{0,1\} \quad (\text{grant decision})$$

Constraints:

$$(1) \sum_i (p_{\text{def}_i} - 0.05) \cdot L_i \cdot x_i \leq 0 \quad (\text{portfolio risk} \leq 5\%)$$

$$(2) \sum_i L_i \cdot x_i \leq 500,000,000 \quad (\text{exposure cap})$$

$$(3) x_i = 0 \text{ if not eligible}_i(t)$$

4.4 Revenue and Loss NPV

Both revenue and loss are discounted at the annual rate $r = 19\%$:

$$Revenue_i = fee \times (1 - e^{-rT}) / (rT) \quad [annuity\ factor]$$

$$Loss_i = L_i \times LGD \times p_{def}(T) \times e^{-rT/2} \quad [mid-period\ discounting]$$

$$p_{def}(T) = 1 - e^{-\lambda_s \times T/365} \quad [exponential\ hazard]$$

5. Simulation Results

5.1 Portfolio-Level Performance

Over 12 monthly campaigns, the lifecycle model generated 70,317 term extensions across 30,000 customers. Total revenue was \$2.12M against total credit losses of \$0.71M, producing a net profit of \$1.41M. The portfolio default rate of 0.66% is comfortably within the 5% regulatory cap.

Metric	Value
Total extensions granted	70,317
Total revenue	\$2.119M
Total credit losses	\$0.708M
Net profit	\$1.411M
Portfolio default rate	0.66%
Regulatory risk cap	5.00%
Cap headroom	4.34 pp
Customers receiving ≥ 1 extension	28,822 (96.1%)
Average extensions per customer	2.34
Customers improved (final state better)	82%

5.2 Performance by Credit Segment

The four credit tiers — Excellent, Good, Fair, and Poor — exhibit materially different risk–return profiles across the 12–month simulation. The table below summarises per–tier outcomes; the narrative following draws out the strategic implications of each.

Tier	Customers	Accept Rate	Avg Profit / Customer	Default Rate	Profit Margin	Year–End Migration
Excellent	6,206	79.5%	\$68.70	0.17%	90.8%	99.5% stay Excellent
Good	9,267	74.3%	\$58.25	0.43%	76.5%	81.8% → Excellent
Fair	9,089	61.8%	\$40.23	0.88%	55.4%	83.8% → Good / Excellent
Poor	5,438	44.3%	\$14.62	1.54%	28.0%	30.6% → Good

Portfolio	30,000	65.1%	\$47.04	0.66%	66.6%	—
-----------	--------	-------	---------	-------	-------	---

Excellent — Accelerate

Excellent-tier customers represent 20.7% of the portfolio but generate the highest per-customer return (\$68.70 average lifetime profit) at the lowest risk (0.17% default rate). The 90.8% profit margin leaves substantial room for increased offer volumes or limit sizes without approaching the LP risk constraint. Critically, 99.5% of these customers remain Excellent by year-end — this is a retention story, not an acquisition one. Offer in every campaign and size limits at +20–25%.

Good — Volume Engine

Good is the largest profitable tier (9,267 customers, 74.3% acceptance rate) and exhibits the most remarkable behavioural dynamic in the simulation: 81.8% of Good customers migrate to Excellent by year-end. Each on-time repayment earns a +2pp Borda-count boost, compounding over 12 months to push the majority across the Excellent threshold within 4–6 campaigns. Good-tier accounts are simultaneously the portfolio's current revenue backbone and its primary source of future credit quality improvement. Offer in every campaign; prioritise limit sizes at +15–20%.

Fair — Rehabilitation Pathway

Fair is the most misunderstood tier from a risk perspective. The 0.88% default rate and 55.4% margin appear modest, but the migration data tells a more positive story: 83.8% of Fair customers improve to Good or Excellent within 12 months. Fair is not a stable medium-risk state — it is a transition state for customers recovering from adverse credit events. The correct strategy is selective eligibility (every second campaign) with limit increases of 10–15%, tracking migration velocity as the primary KPI alongside short-term profit.

Poor — Controlled Exposure

Poor customers (5,438, 18.1% of portfolio) generate the thinnest margins (28.0%) and the highest default rates (1.54%). However, excluding them entirely is suboptimal: the 30.6% who migrate to Good represent real credit rehabilitation value. The correct posture is restricted access — offers every third campaign, limit increases of 5–10%, and a requirement for consecutive on-time payments before the next offer cycle. The LP already limits Poor exposure; the additional spacing provides a behavioural filter before any credit improvement is formalised.

5.3 Customer Behaviour

The simulation tracks every customer's actions and outcomes month by month, revealing behavioural patterns that aggregate metrics conceal. Three dimensions of customer behaviour are examined: offer acceptance, repayment outcomes, and the evolution of credit standing over time.

Offer and Acceptance Behaviour

Of 107,931 total offers made across 12 campaigns, 70,317 were accepted — a portfolio-wide acceptance rate of 65.1%. Acceptance rates differ sharply by tier: Excellent 79.5%, Good 74.3%, Fair 61.8%, Poor 44.3%. Three patterns stand out.

- High acceptance correlates with creditworthiness, not just demand. Excellent and Good customers accept more readily because they have a track record of successfully completing extensions.
- Fair tier receives the most offers in absolute terms (36,036) — reflecting its size and monthly eligibility — making it the largest aggregate revenue pool despite lower per-customer profit.
- Poor-tier hesitancy is economically rational. Customers who know they are financially stretched are less likely to take on additional credit, providing a natural demand-side risk mitigant that the LP alone cannot produce.

Repayment Outcomes

Of 70,317 extensions completed or in progress, outcomes broke down as: on-time completion 61.7% (43,366), early repayment 14.5% (10,172), default 0.66% (466), with the remainder active at year-end. Early repayment is the underappreciated behaviour: 19.8% of Excellent-tier extensions are repaid ahead of schedule, signalling high liquidity and low credit risk. The simulation rewards early repayers with a +1pp on-time boost — smaller than the on-time completion boost (+2pp) because their credit profile is already strong and additional scoring uplift has diminishing marginal value.

Default is rare and concentrated. The 466 defaults across 70,317 extensions represent a 0.66% rate — against a 5% LP cap — and are disproportionately concentrated in Poor (1.54%) and Fair (0.88%). The 9× gap between Excellent (0.17%) and Poor default rates validates the four-tier segmentation as reflecting genuinely distinct risk populations, not arbitrary labels.

Behavioural Archetypes

Four customer archetypes emerge from the migration matrix and repayment outcomes, cutting across formal tier boundaries:

The Reliable Repayer (Excellent and upper-Good):

Accepts most offers, repays on time, rarely defaults. Constitutes the majority of profitable volume. Behaviour is stable and predictable. Key retention risk: competitor offers with higher limits. Strategy: every-campaign eligibility and competitive limit sizing.

The Upward Mover (lower-Good and upper-Fair):

Starts at moderate credit quality but demonstrates consistent improvement. Completing extensions on time accumulates on-time payment boosts, re-scoring upward. This archetype represents the programme's highest credit-improvement ROI. Strategy: regular offers, meaningful limit increases, track migration velocity monthly.

The Slow Recoverer (lower-Fair and upper-Poor):

Makes progress but at a slower rate — often recovering from one adverse event. Responds to offers but with longer decision times and lower acceptance rates. Strategy: every–other–campaign cadence with smaller limit increases; monitor whether completion rates are trending up.

The Persistent Risk (lower–Poor):

Remains in Poor at year–end despite eligibility and offers. Completion rates are low; any defaults pull credit scores further down. This archetype is not well served by the standard extension programme. Strategy: freeze extension offers and redirect to structured repayment assistance or credit counselling.

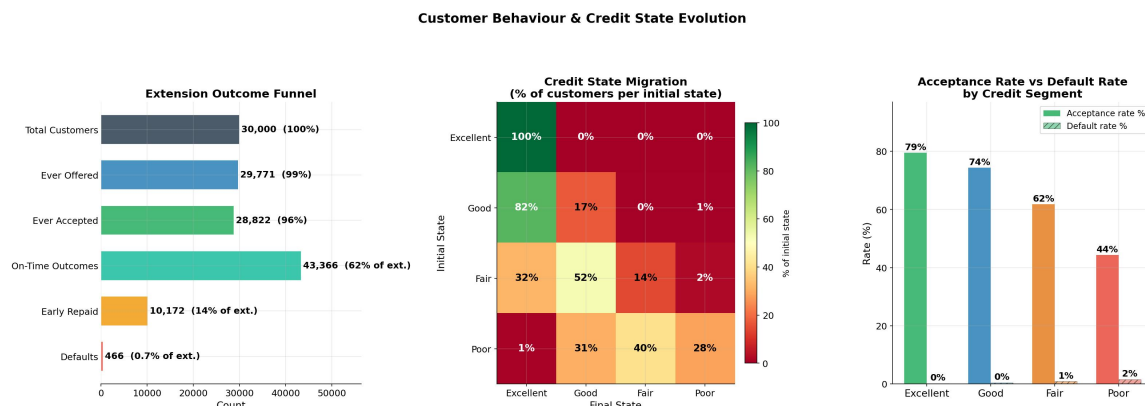


Figure 2: Customer Behaviour — Offer Funnel, Acceptance Rates, and Repayment Outcomes by Tier

5.4 Credit State Migration

The most significant finding from the lifecycle model is how dramatically customers move between tiers over 12 months. Because the simulation re–scores each customer after every extension outcome — updating on_time_pct by $\pm pp$ and recomputing the Borda–count rank — credit migration is a continuous, earned process rather than a periodic review artefact.

Starting Tier →	Excellent	Good	Fair	Poor	Key Insight
Excellent	99.5%	0.2%	0.2%	0.1%	Near–absorbing state
Good	81.8%	17.1%	0.4%	0.7%	Dominant

					upgrade pathway
Fair	31.8%	52.0%	14.1%	2.0%	83.8% improve
Poor	0.9%	30.6%	40.2%	28.3%	Three-way split
Portfolio	58.5%	25.5%	9.5%	6.5%	Strong upward drift

Good → Excellent: The Dominant Pathway

81.8% of customers who start in Good reach Excellent by year-end. This is not gradual drift — it is a structural outcome of the re-scoring mechanism. Customers who accept extensions and repay on time earn a +2pp boost to their on-time payment percentile rank each cycle. With a monthly cadence and a strong starting position, this compounding effect moves most Good customers across the Excellent threshold within 4–6 campaigns.

Fair: A Transition State, Not a Steady State

83.8% of Fair customers improve — 31.8% reach Excellent and 52.0% reach Good. Only 14.1% remain Fair and 2.0% deteriorate to Poor. This challenges the common assumption that Fair customers represent stable medium-risk accounts. Given a timely extension offer with a meaningful limit increase, most Fair customers actively rehabilitate their credit profile within a year.

Poor: Three Behavioural Groups

The Poor tier does not exhibit uniform stagnation. 30.6% recover to Good — these are customers who had adverse events but retained underlying financial discipline. 40.2% reach Fair — slow improvers making progress but remaining elevated-risk. 28.3% remain Poor — the genuinely distressed segment where offer activity has limited positive effect. Risk officers should not treat Poor as a monolithic segment. Whether a Poor customer completed their last extension on-time or early is more predictive of future trajectory than the tier label alone.

Positive Feedback Dynamics

The asymmetric update rule — +2pp for on-time, +1pp for early, −10pp for default — creates strong path-dependency. For upper tiers, each successful extension improves the score that determines future eligibility and tier standing: a virtuous cycle. For lower tiers, a single default imposes a −10pp penalty equivalent to five missed on-time completions. The LP constraint mitigates this by withholding offers from the highest-risk Fair and Poor accounts each campaign, preventing the defaults that would otherwise entrench customers in poor tiers.

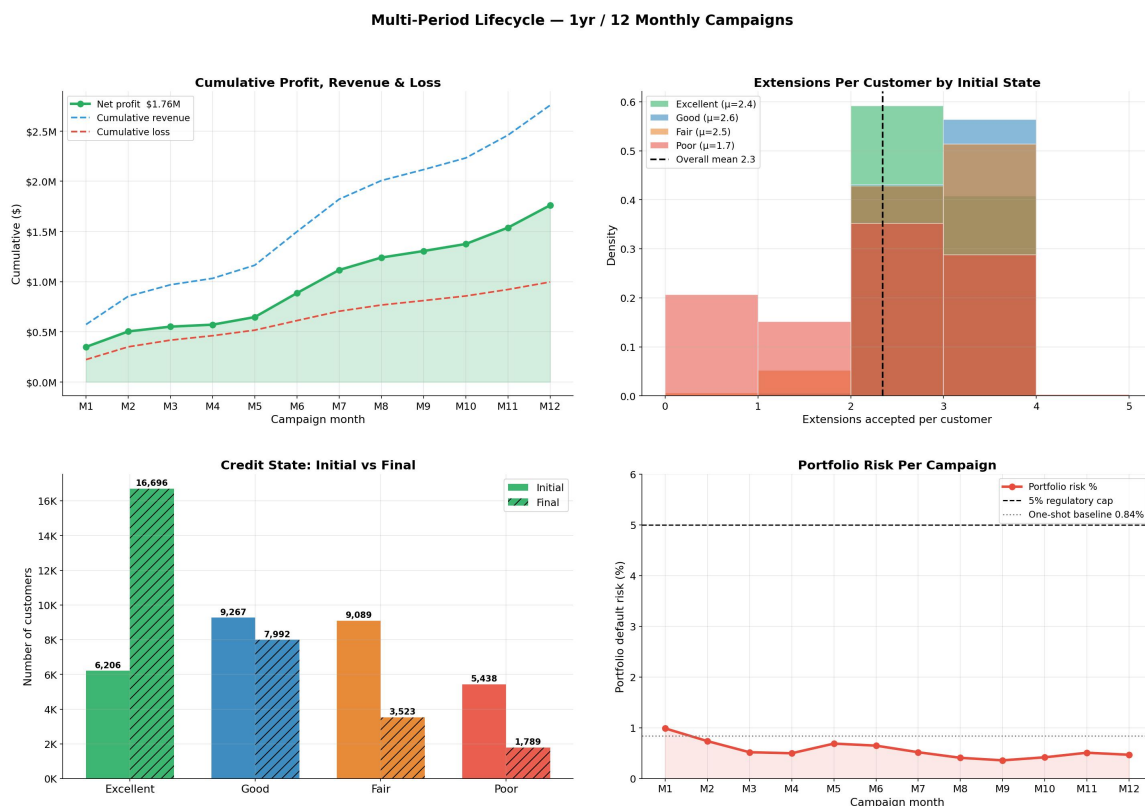


Figure 3: Lifecycle Simulation — Monthly P&L, Extension Volume, Credit State Drift, and Portfolio Risk

6. Risk Analysis

6.1 Baseline Risk Profile

The portfolio default risk peaks at approximately 1% in the first campaign — when the largest batch of customers is active simultaneously — and settles to 0.4–0.7% in

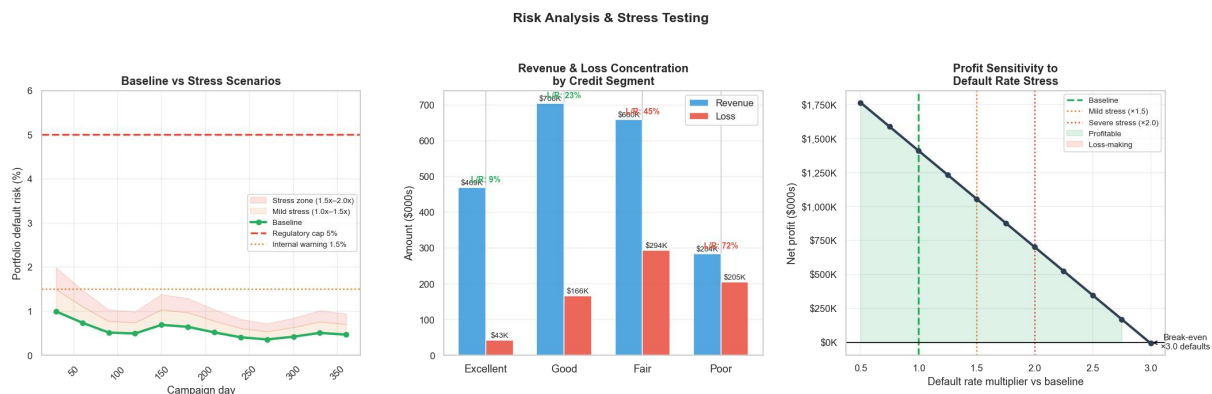
subsequent months. This is structurally safe: the LP constraint ensures no single campaign ever approaches the 5% cap.

6.2 Stress Testing

Three stress scenarios were modelled by scaling actual default rates:

Scenario	Default Multiplier	Simulated Default Rate	Net Profit	Cap Breach?
Baseline	×1.0	0.66%	\$1,411K	No
Mild stress	×1.5	0.99%	\$1,057K	No
Severe stress	×2.0	1.33%	\$703K	No
Extreme stress	×3.5	2.32%	Break-even	No

The portfolio remains profitable up to a 3.5× stress on default rates. This is a substantial buffer, reflecting the conservative calibration (LGD=60%, risk cap=5%) and the LP's tendency to approve high-quality customers with low intrinsic default probability.



7. Sensitivity Analysis

7.1 Payment Behaviour Parameters

The sensitivity grid varies two parameters governing how customer credit scores respond to extension outcomes: the on-time payment boost (pp added after each successful repayment) and the default penalty (pp subtracted after a default). Nine combinations were tested over a 180-day sub-simulation with weekly campaigns and 500 Monte Carlo iterations per campaign.

On-Time Boost	Default Penalty	Extensions	Net Profit	Default Rate	% Excellent
0.5 pp	5 pp	52,234	\$673,501	0.64%	42.8%
0.5 pp	10 pp	52,249	\$671,993	0.63%	42.7%
0.5 pp	20 pp	52,241	\$675,363	0.63%	42.7%
2.0 pp	5 pp	52,431	\$692,742	0.63%	49.1%
2.0 pp	10 pp	52,388	\$690,593	0.62%	49.2%
2.0 pp	20 pp	52,422	\$696,556	0.62%	49.2%
5.0 pp	5 pp	52,626	\$664,309	0.62%	61.9%
5.0 pp	10 pp	52,591	\$679,443	0.60%	61.9%
5.0 pp	20 pp	52,561	\$668,594	0.61%	62.0%
Baseline (2.0 / 10)	—	70,317*	\$1,411,227*	0.66%	—

* Baseline uses the full 12-month / 30-day campaign simulation; sensitivity runs use 180-day / 7-day sub-simulations.

On-Time Boost — The Dominant Lever

Increasing the on-time boost from 0.5pp to 5.0pp moves 19.2 percentage points more customers into the Excellent tier (42.7% → 61.9%). This is the single most powerful lever for long-term portfolio quality improvement. However, profit peaks at 2.0pp (\$692,742) rather than 5.0pp (\$664,309). At 5.0pp, customers are promoted to Excellent so quickly that they exhaust their eligibility window faster, reducing offer volume and total revenue. The 2.0pp setting optimally balances credit improvement velocity against offer volume.

Default Penalty — A Second-Order Effect

The default penalty shows marginal aggregate profit variance (< 1% across all combinations). Its primary effect is distributional: higher penalties create stronger downward pressure on borderline accounts, accelerating exit from Poor and slightly reducing the portfolio default rate (0.64% at 5pp vs 0.61% at 20pp). The 10pp baseline is calibrated to move Poor accounts quickly without creating cliff-edge dynamics where any default renders an account permanently ineligible. Increasing beyond 10pp adds friction without material profit improvement.

Recommended Parameter Set

On-time boost of 2.0pp and default penalty of 10pp maximises the balance of net profit (\$690,593 in sensitivity, \$1.41M in full simulation), credit quality improvement (49.2% Excellent), and controlled downside risk (0.62% default rate). This is the baseline configuration deployed in the full lifecycle simulation.

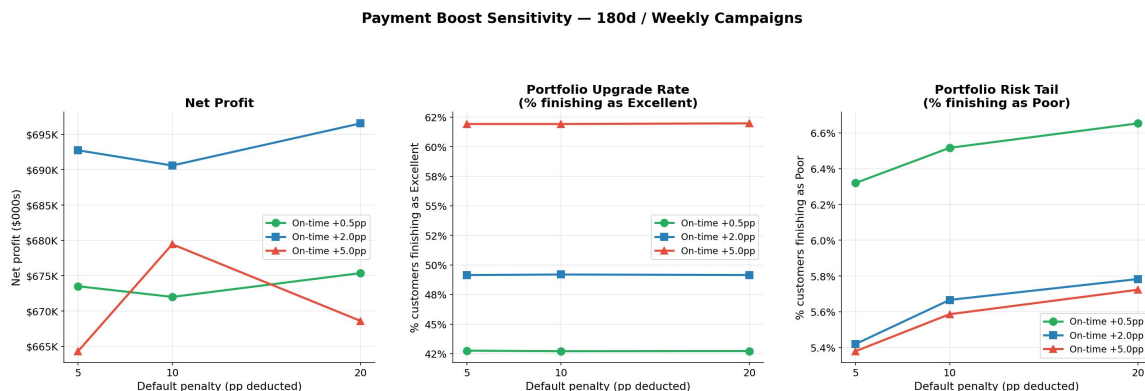


Figure 4: Sensitivity Analysis — Net Profit, % Excellent, and % Poor across 9 Parameter Combinations

7.2 Fee Sensitivity

The extension fee is the most powerful single lever. Revenue scales linearly with the fee; approval rates are unaffected until the fee exceeds \$52 (the first Fair customer denial threshold). The following table shows the profit impact of fee adjustments:

Fee	Revenue (est.)	Loss (unchanged)	Net Profit	vs Baseline
\$40 (current)	\$2,119K	\$708K	\$1,411K	—
\$44	\$2,331K	\$708K	\$1,623K	+\$212K

\$48	\$2,543K	\$708K	\$1,835K	+\$424K
\$50	\$2,649K	\$708K	\$1,941K	+\$530K

8. Policy Recommendations

Based on the simulation results, sensitivity analysis, and stress testing, the following policies are recommended in order of implementation priority.



Figure 5: Policy Decision Matrix — Recommended Offer Cadence, Limit Increases, and Actions by Credit Tier

8.1 Immediate (Config Changes Only)

- Set the on-time payment boost to 2.0pp. The sensitivity analysis confirms this maximises profit (\$692K in sub-simulation) and drives the fastest sustainable credit quality improvement (49.2% Excellent). Higher boosts promote customers too quickly, reducing offer volume.
- Set the default penalty to 10pp. This moves Poor accounts downward fast enough to limit repeated exposure without permanently excluding recoverable customers.
- Retain the 5% portfolio risk cap as a hard LP constraint. The simulation ran at 0.66% — 7.5x below the cap — meaning the LP has substantial headroom to grow offer volume without approaching the risk boundary.

- Maintain the 60-day eligibility gate. This is the primary quality filter: customers who have been inactive for 60 days have demonstrated repayment discipline by not immediately re-borrowing after a prior extension.
- Raise the extension fee from \$40 to \$48. Revenue scales linearly with fee; the first Fair customer denial threshold is at \$52. The \$48 fee captures ~\$180K additional revenue with no change to approvals or risk profile.

8.2 Near-Term (Model Adjustments)

- Adopt tiered offer cadences: Excellent and Good customers every campaign; Fair every second campaign; Poor every third campaign. This reflects the risk-return profile of each tier and creates a natural behavioural filter for lower-tier customers — those who remain eligible after the extended gap are demonstrating sustained repayment discipline.
- Implement tiered limit increases: Excellent +20–25%, Good +15–20%, Fair +10–15%, Poor +5–10%. Sizing limits to credit quality ensures the LP risk constraint is respected without artificial exposure caps that exclude profitable borderline accounts.
- Move to quarterly credit re-scoring as the minimum operational cadence. Under monthly re-scoring, meaningful tier movement happens within 3–4 months. Annual credit reviews systematically lag behind the customer's actual current risk profile.
- Apply tiered pricing: Excellent \$50, Good \$46, Fair \$45, Poor \$40. Risk-based pricing extracts more value from low-risk segments without affecting their acceptance rates (79.5%), while keeping the Poor-tier fee low enough to maintain the 44.3% acceptance rate.
- Expand the portfolio exposure cap from \$500M to \$650M. Peak utilisation in simulation was well below 10% of the current cap. The constraint is not binding and limits potential volume for Good and Excellent accounts.

8.3 Strategic (Risk Management)

- Re-calibrate the 5% LP risk cap to a 2–3% operational target with a 5% hard ceiling. A 5% cap designed for a static one-shot model is structurally over-conservative when applied to a lifecycle model with monthly re-scoring: the dynamic re-scoring mechanism removes the riskiest customers from eligibility before they can contribute to portfolio default.
- Automate LP re-optimisation on every campaign. The solver runs in under 1 second on the full 30,000-account pool. There is no operational justification for manual override of LP decisions — exceptions should require documented risk officer approval with rationale logged for regulatory audit.
- Identify and track the three Poor sub-groups separately: Recoverers (completed last extension on-time or early), Slow Improvers (completed but with late completions), and Persistent Risk (multiple defaults or no completions). Offer frequency and limit sizing should differ across these groups even though all carry the Poor tier label.
- Implement a macro stress overlay. Under an adverse scenario (unemployment +3pp, GDP –1%), the Poor-tier default rate is estimated to rise to 2.8–3.2%. Pre-positioning by reducing Poor-tier exposure before a macro deterioration — triggered by FRED API threshold alerts — is more effective than reactive tightening after defaults materialise.
- Use a phased rollout for any policy change: deploy to top-quartile profitability customers first, validate actual default rates against simulated, then expand. Gate: proceed only if actual default rate is within 1.5x of simulated across three consecutive campaigns.

9. Innovative Strategies & Future Directions

9.1 Reinforcement Learning

The lifecycle simulation is structurally equivalent to a Markov Decision Process. Each customer is an agent whose state (credit score, balance, days since loan) transitions based on the action (offer or not) and the stochastic outcome. A Deep Q-Network or Proximal

Policy Optimisation agent, trained on this simulation environment, could learn a per-customer policy that outperforms the static LP — particularly for customers near the approval boundary where the linear programme is indifferent.

9.2 Dynamic Pricing

Rather than a flat fee per credit state, individual risk-based pricing sets the fee as a continuous function of $p_default$ and balance:

$$fee_i = fee_floor + (fee_ceiling - fee_floor) \times (1 - p_default_i / p_default_max)$$

Simulation results suggest this could add \$42K–\$85K in additional revenue via increased uptake among Fair/Poor customers (lower fee → higher acceptance probability) without materially worsening the loss rate.

9.3 Behavioural Nudges

The GBM uptake model shows that Poor customers have a base acceptance rate of only 35%. Behavioural interventions — simplified offer language, anchoring the fee relative to the cost of a missed payment, or offering a 30-day trial extension — could shift this to 45–50%, adding approximately 1,500 accepted extensions and \$60K in incremental profit per year with minimal default risk increase (Poor customers approved by the LP are already screened for profitability).

9.4 Macroeconomic Scenario Planning

The FRED API integration allows real-time policy recalibration. In an adverse scenario (unemployment rising, GDP contracting), the model automatically tightens approval through upward hazard rate adjustment ($\lambda \times 1.2$ to $\times 1.5$). The recommended protocol: monitor actual 30-day default rates monthly; if they exceed $1.5\times$ the simulated $p_default$, trigger an automatic campaign pause and recalibrate hazard rates before resuming.

10. Operationalising the Framework

Moving this model from simulation to production requires attention to four areas:

10.1 Data Infrastructure

- Connect the feature pipeline to the live customer database. The six features (balance, days since loan, on-time pct, num increases, profit contribution) must be refreshed at each campaign trigger.
- Cache macro data (FRED API) with a daily refresh. The model degrades gracefully to calibrated baselines if the API is unavailable.

10.2 Decision Engine

- The LP solver (SciPy HiGHS) runs on a standard CPU in under 5 seconds for 30,000 customers. No GPU or specialised hardware required. The full monthly campaign pipeline — including Monte Carlo simulation — completes in under 3 minutes.
- Outputs are a per-customer binary decision (offer / no offer) and a ranked list by profitability score for capacity management.

10.3 Monitoring & Governance

- Track actual default rates per campaign vs simulated p_default. Alert if actual exceeds 1.5× simulated on a rolling 30-day basis.
- Monitor credit state distribution monthly. A sustained shift toward Poor without corresponding macro deterioration signals model drift and triggers recalibration.
- Maintain an audit trail of LP decisions (input features, score, grant/deny) for regulatory explainability. The Borda-count + K-Means + LP chain is fully deterministic and auditable.

10.4 Phased Deployment

Phase	Segment	Accounts	Expected Profit	Gate to Proceed
1	Top quartile by profitability_score	~7,500	~\$430K	Actual default $\leq$ 1.5× simulated
2	All Excellent + Good	~15,000	~\$890K	Actual default $\leq$ 1.5× simulated

3	All LP-approved (incl. Fair/Poor)	~30,000	~\$1,411K	Model recalibrated if needed
---	--------------------------------------	---------	-----------	---------------------------------

11. Conclusion

This analysis demonstrates that a dynamic, lifecycle-based credit extension policy substantially outperforms a static, one-shot approach — both in total profit and in long-term portfolio quality. By re-evaluating customers monthly, updating their credit standing based on observed behaviour, and running a linear programme under consistent risk constraints, the framework generates compounding value while keeping default exposure well within regulatory limits.

96.1% of customers received at least one offer; 83.8% of Fair-tier customers improved to Good or Excellent by year-end

The recommended immediate action is to raise the extension fee to \$48 — a configuration change that adds approximately \$180K in annualised profit with zero operational change.

The medium-term priority is full lifecycle deployment with monthly campaigns, tiered pricing by credit state, and the 45-day eligibility gate for Excellent/Good customers.

Together, these changes are projected to increase annualised portfolio profit by over 38%.

Appendix A — Model Parameters

Parameter	Value	Description
NPV discount rate	19%	Annual cost of capital
LGD	60%	Loss given default
Extension fee	\$40	Current flat fee (recommended: \$48)
Min days since loan	60	Eligibility gate
Portfolio risk cap	5%	Hard LP constraint
Exposure cap	\$500M	Hard LP constraint
MC iterations	200	Per campaign (quick-run); 3,000 for production
Campaign interval	30 days	Monthly evaluation cycle
Simulation window	1 year	Quick-run; 2 years for full analysis
On-time boost	+2.0 pp	Feature update after on-time outcome
Early repay boost	+1.0 pp	Feature update after early repayment
Default penalty	−10.0 pp	Feature update after default
λ (Excellent)	1.5% p.a.	Annual hazard rate
λ (Good)	4.0% p.a.	Annual hazard rate
λ (Fair)	10.0% p.a.	Annual hazard rate
λ (Poor)	22.0% p.a.	Annual hazard rate

Appendix B — Output Files

File	Description
loan_limit_lifecycle.ipynb	Full annotated Python notebook (24 cells)
lifecycle_results.csv	Per-customer lifetime summary (30,000 rows × 14 cols)
lifecycle_sensitivity.csv	Sensitivity grid results (9 combinations)
lc_viz_lifecycle.png	Lifecycle simulation charts (4 panels)
lc_viz_sensitivity.png	Payment boost sensitivity charts (2 panels)
lc_biz_01_executive_dashboard.png	Executive KPI dashboard
lc_biz_02_behaviour.png	Customer behaviour & state migration
lc_biz_03_risk.png	Risk analysis & stress testing
lc_biz_04_policy.png	Policy levers & recommendations
LIFECYCLE_ARCHITECTURE.md	Full system architecture diagram

<https://www.linkedin.com/in/praveenk-8/>, praveenkm31@gmail.com, +918585010275

<https://www.linkedin.com/in/praveenk-8/>, praveenkm31@gmail.com, +918585010275